

SW_PWR_PCB
NRST
UART_TX_POD1
UART_RX_POD1
MAG_SW
PWM_M7
Analog_M2
PWM_M6
Analog_M1
PWM_M5
PWM_M4
Analog_M8
PWM_EXT_LED
Analog_M7
Analog_M6
PWM_MTL
Analog_M5
SW_V_BAT
Analog_M4
I2C_SDA_1
I2C_SCL_1
LED_STRIP_1
LED_STRIP_2
JTAG_SM
JTAG_CK
JTAG_DI
PWM_TORP
Analog_M3
PWM_M8
JTAG_DO
JTAG_RST
PWM_M3
PWM_M2
PWM_M1
PWM_GRIPPER
SW_3V3
SW_5V
SW_12V
I2C_SCL_3
I2C_SDA_3
I2C_SCL_2
I2C_SDA_2
UART_TX_EXP2
UART_RX_EXP2
UART_TX_EXP1
UART_RX_EXP1
UART_TX_POD2
UART_RX_POD2
UART_TX_SCRN
UART_RX_SCRN
ADC_TEMP_1
ADC_TEMP_2
I2C_SCL_4
I2C_SDA_4
I2C_SDA_5
ALERT_TEMP
SIGNAL_LEAK
I2C_SCL_5
BOOT0_JUMP
GPIO429

U_MCU
MCU.SchDoc

SW_PWR_PCB
NRST
UART_TX_POD1
UART_RX_POD1
MAG_SW
PWM_M7
Analog_M2
PWM_M6
Analog_M1
PWM_M5
PWM_M4
Analog_M8
PWM_EXT_LED
Analog_M7
Analog_M6
PWM_MTL
Analog_M5
SW_V_BAT
Analog_M4
I2C_SDA_1
I2C_SCL_1
LED_STRIP_1
LED_STRIP_2
JTAG_SM
JTAG_CK
JTAG_DI
PWM_TORP
Analog_M3
PWM_M8
JTAG_DO
JTAG_RST
PWM_M3
PWM_M2
PWM_M1
PWM_GRIPPER
SW_3V3
SW_5V
SW_12V
I2C_SCL_3
I2C_SDA_3
I2C_SCL_2
I2C_SDA_2
UART_TX_EXP2
UART_RX_EXP2
UART_TX_EXP1
UART_RX_EXP1
UART_TX_POD2
UART_RX_POD2
UART_TX_SCRN
UART_RX_SCRN
ADC_TEMP_1
ADC_TEMP_2
I2C_SCL_4
I2C_SDA_4
I2C_SDA_5
ALERT_TEMP
SIGNAL_LEAK
I2C_SCL_5
BOOT0_JUMP
GPIO429

PWM_M1
PWM_M2
PWM_M3
PWM_M4
PWM_M5
PWM_M6
PWM_M7
PWM_M8
PWM_TORP
Analog_M1
Analog_M2
Analog_M3
Analog_M4
Analog_M5
Analog_M6
Analog_M7
Analog_M8
PWM_GRIPPER
PWM_EXT_LED
PWM_MTL
UART_TX_SCRN_5V
UART_RX_SCRN_5V
LEAK_PROBES
LED_STRIP_1
ADC_TEMP_1
JTAG_RST
JTAG_DI
JTAG_SM
JTAG_CK
JTAG_DO
NRST
I2C_SDA_1
I2C_SCL_1
I2C_SDA_4
I2C_SCL_4
UART_TX_XAV
UART_RX_XAV
UART_TX_XAV_5V
UART_RX_XAV_5V
UART_TX_POD1
UART_RX_POD1
UART_TX_POD2
UART_RX_POD2
UART_TX_EXP1_5V
UART_RX_EXP1_5V
UART_TX_EXP2
UART_RX_EXP2
SW_3V3
SW_12V
SW_PWR_PCB
SW_5V
SW_V_BAT
MAG_SW
ADC_TEMP_2
LED_STRIP_2
I2C_SCL_2
I2C_SDA_2
I2C_SCL_3
I2C_SDA_3
GPIO429
I2C_SCL_5
BOOT0_JUMP

U_CONNECTORS
CONNECTORS.SchDoc

PWM_M1
PWM_M2
PWM_M3
PWM_M4
PWM_M5
PWM_M6
PWM_M7
PWM_M8
PWM_TORP
Analog_M1
Analog_M2
Analog_M3
Analog_M4
Analog_M5
Analog_M6
Analog_M7
Analog_M8
PWM_GRIPPER
PWM_EXT_LED
PWM_MTL
UART_TX_SCRN_5V
UART_RX_SCRN_5V
LEAK_PROBES
LED_STRIP_1
ADC_TEMP_1
JTAG_RST
JTAG_DI
JTAG_SM
JTAG_CK
JTAG_DO
NRST
I2C_SDA_1
I2C_SCL_1
I2C_SDA_4
I2C_SCL_4
UART_TX_XAV
UART_RX_XAV
UART_TX_XAV_5V
UART_RX_XAV_5V
UART_TX_POD1
UART_RX_POD1
UART_TX_POD2
UART_RX_POD2
UART_TX_EXP1_5V
UART_RX_EXP1_5V
UART_TX_EXP2
UART_RX_EXP2
SW_3V3
SW_12V
SW_PWR_PCB
SW_5V
SW_V_BAT
MAG_SW
ADC_TEMP_2
LED_STRIP_2
I2C_SCL_2
I2C_SDA_2
I2C_SCL_3
I2C_SDA_3
GPIO429
I2C_SCL_5
BOOT0_JUMP

UART_TX_SCRN
UART_TX_SCRN_5V
UART_RX_SCRN_5V
UART_RX_SCRN
UART_RX_XAV
UART_TX_XAV
UART_RX_XAV_5V
UART_TX_XAV_5V
UART_RX_EXP1
UART_TX_EXP1
UART_RX_EXP1_5V
UART_TX_EXP1_5V

U_LEVEL_SHIFTERS
LEVEL_SHIFTERS.SchDoc

UART_TX_SCRN
UART_TX_SCRN_5V
UART_RX_SCRN_5V
UART_RX_SCRN
UART_RX_XAV
UART_TX_XAV
UART_RX_XAV_5V
UART_TX_XAV_5V
UART_RX_EXP1
UART_TX_EXP1
UART_RX_EXP1_5V
UART_TX_EXP1_5V

LEAK_PROBES
SIGNAL_LEAK
ALERT_TEMP
I2C_SDA_5
I2C_SCL_5

U_SENSORS
SENSORS.SchDoc

LEAK_PROBES
SIGNAL_LEAK
ALERT_TEMP
I2C_SDA_5
I2C_SCL_5

Title

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8-11-2026

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Revision

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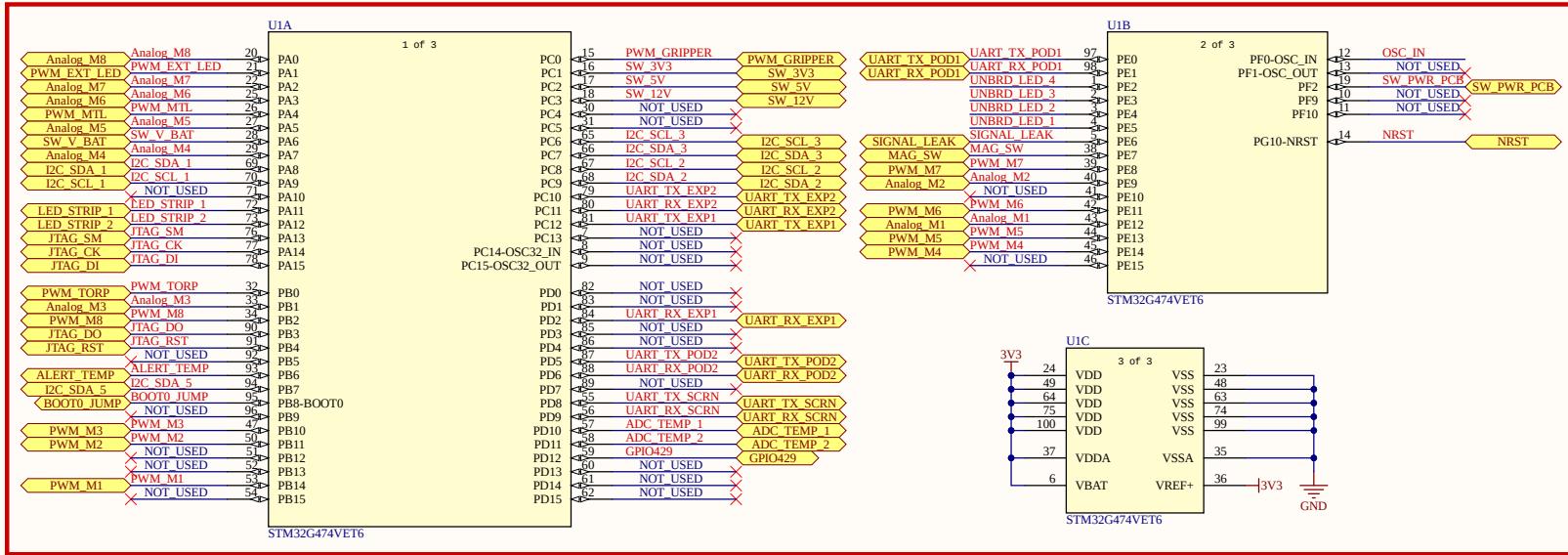
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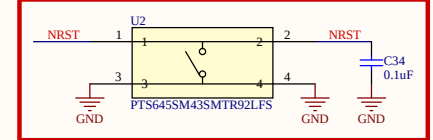
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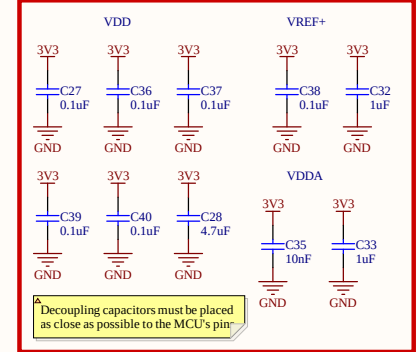
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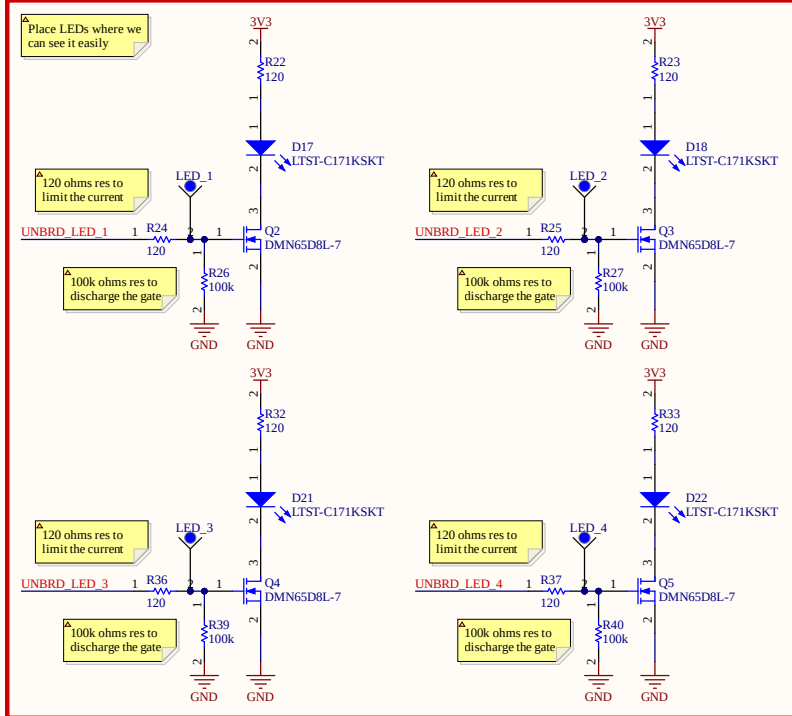
Negative Reset (NRST) Button



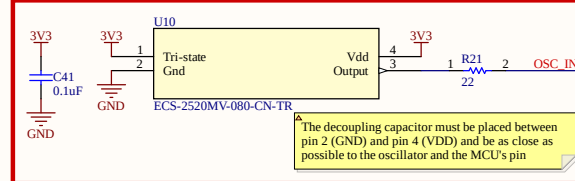
MCU's Decoupling Capacitors



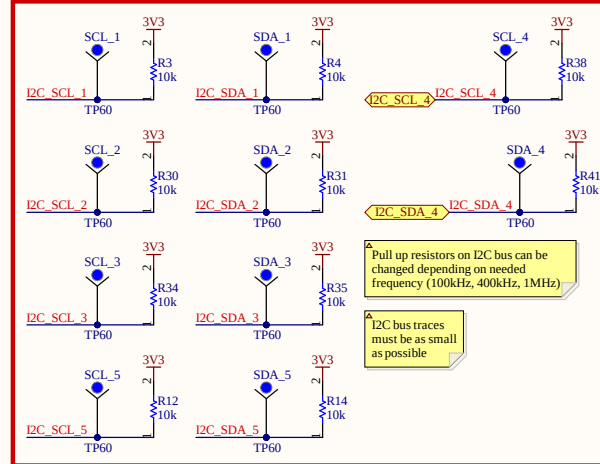
Unboard LEDs (For Debugging)



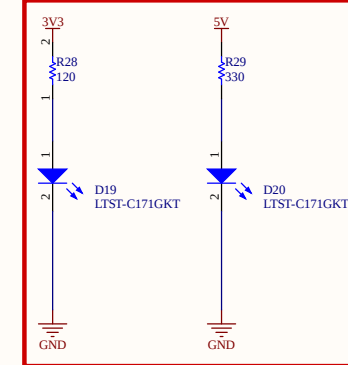
Negative Reset (NRST) Button



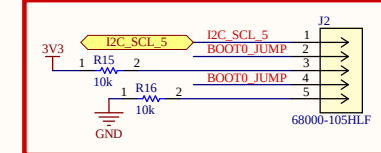
I2C Test Points / Pull Up Resistors



Power LEDs

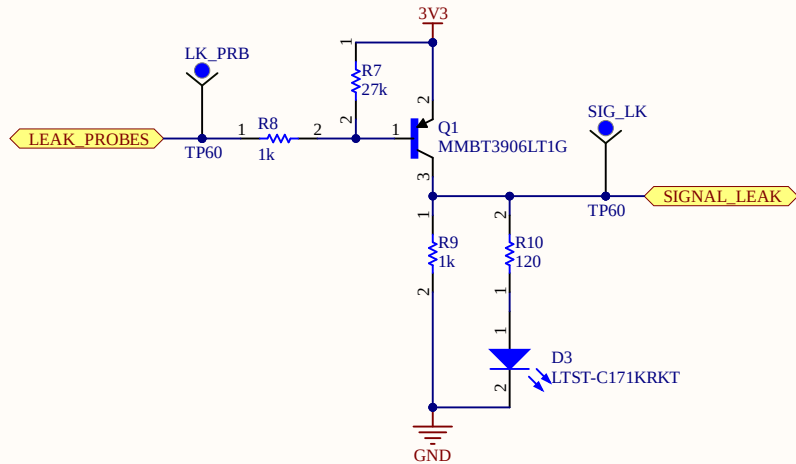


BOOT0 Jumper

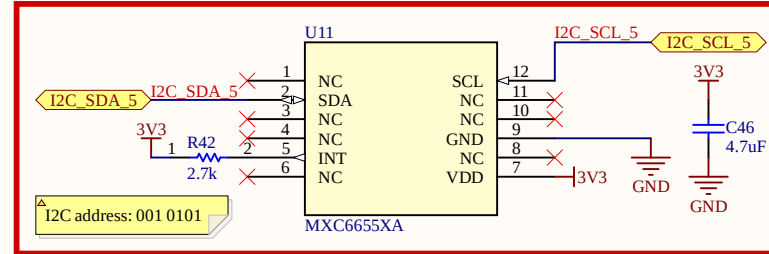


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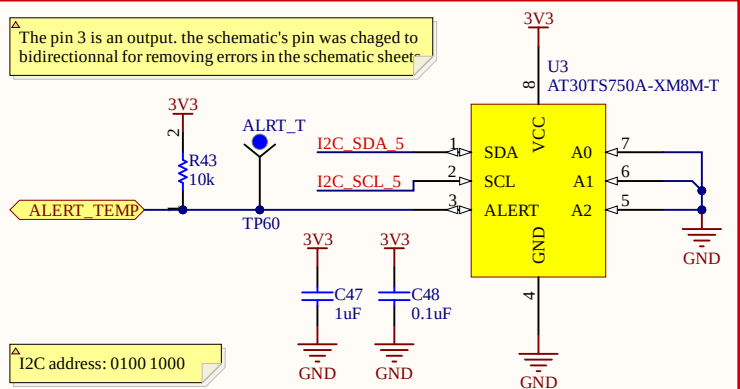
Leak Sensor Circuit



3 Axis Accelerometer

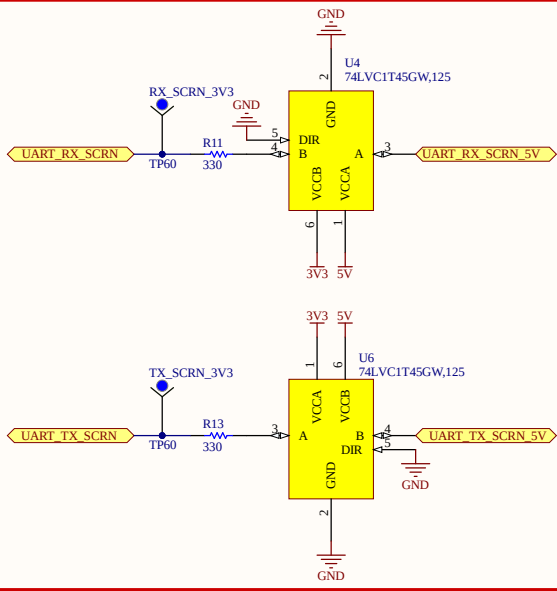


Temperature Sensor

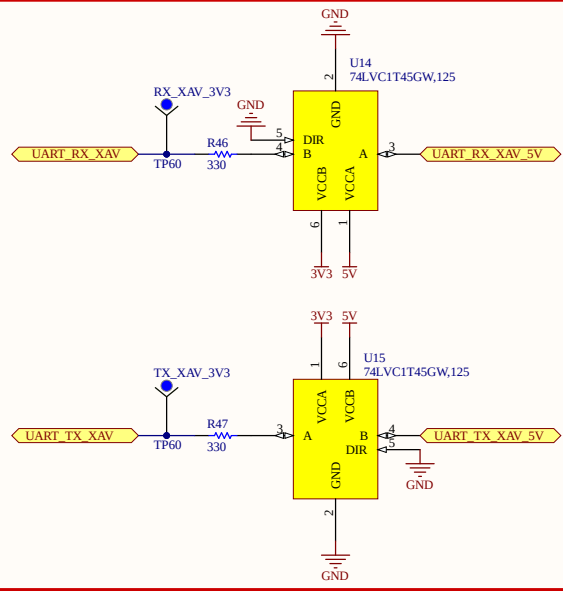


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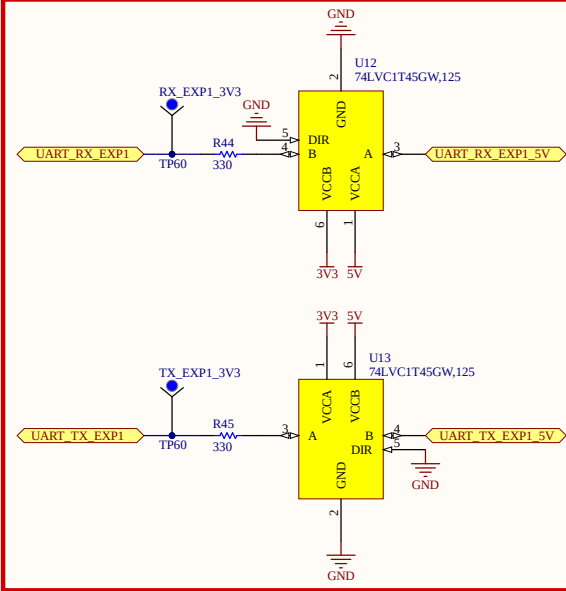
Unboard Screen Level Shifters



UART Xavier Level Shifters (For DVL)



UART Expander 1 Level Shifters



Title		
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